Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

**Vision of the Department**

*To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.*

**Mission of the Department**

*To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.*

**Session 2026-2027(ODD)**

| | |
|---|---|
| **Vision:** Dream of where you want. | **Mission:** Means to achieve Vision |

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

| | | | |
|---|---|---|---|
| PEO1 | **Preparation** | **P: Preparation** | **Pep-CL abbreviation pronounce as Pep-si-lL easy to recall** |
| PEO2 | **Core Competence** | **E: Environment (Learning Environment)** | |
| PEO3 | **Breadth** | **P: Professionalism** | |
| PEO4 | **Professionalism** | **C: Core Competence** | |
| PEO5 | **Learning Environment** | **L: Breadth (Learning in diverse areas)** | |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

**Keywords of POs:**
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Srushti Chaudhari**



**27-09-2026**
**Name and Signature of Student and Date**
(Signature and Date in Handwritten)

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| **Session** | **2026-27 (ODD)** | **Course Name** | **LAB: Software Engineering** |
|---|---|---|---|
| **Semester** | **7th** | **Course Code** | **23CT1704** |
| **Roll No** | **24** | **Name of Student** | **Srushti Chaudhari** |

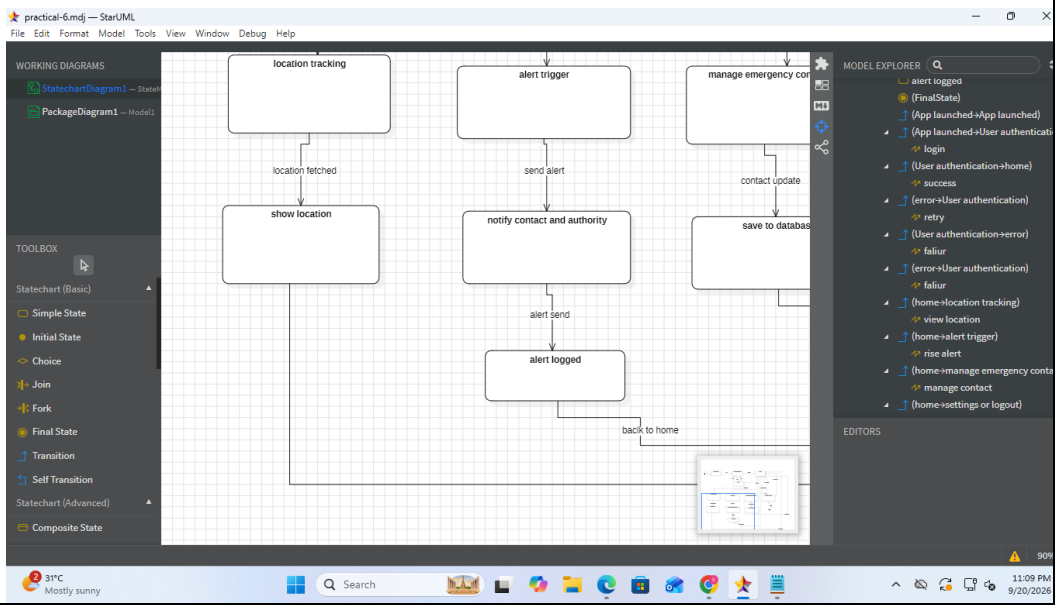
| Practical Number | **7** |
|---|---|
| Course Outcome | Upon successful completion of the course the students will be able to;<br>1. Choose appropriate software engineering process model, requirement engineering principles and software designing fundamentals for a given project.<br>2. Select appropriate testing strategy and apply testing principles for testing a given application.<br>3. Apply basics of software configuration management, version control and change control in software development.<br>4. Evaluate cost estimation, effort and severity of software risk for given application.<br>5. Perform basic operations on Sub-version for software version control. |
| Aim | To study and draw the **State Diagram** for the given case study – **Women Safety Alert System**. |
| Problem Definition | To design a State Diagram for a Women Safety Alert System that shows the different states of the system and the transitions between them. The system allows a user to register or log in, access the home screen, track location, trigger an SOS alert, notify emergency contacts and authorities, manage contacts, and log out of the system. |
| Theory (100 words) | A **State Diagram** is a behavioral UML diagram that shows the different states of a system and the transitions between those states. A state represents a condition or activity of the system, while a transition shows movement from one state to another due to an event or action. In a Women Safety Alert System, the main states include App Launched, Authentication, Home, Location Tracking, Alert Triggered, Notification, Alert Logged, Manage Contacts, and Logout. The diagram helps to understand how the system behaves during normal use and during an |

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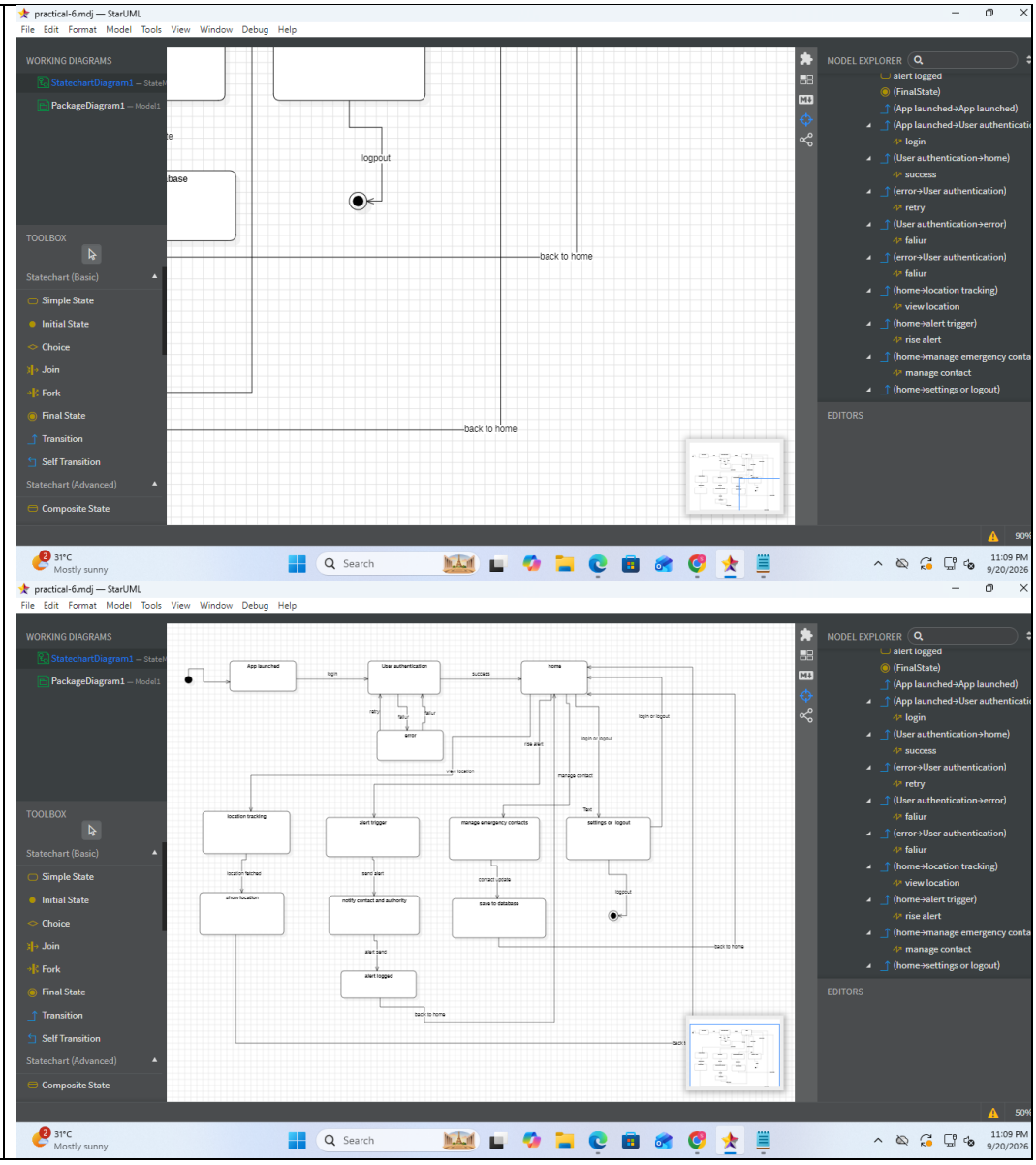
| | |
|---|---|
| | emergency. It provides a simple view of the dynamic behavior of the system.. |
| Procedure and Execution<br><br>(100 Words) | Algorithm(Steps):<br>1. Open **StarUML** software.<br>2. Create a new project.<br>3. Select **Statechart Diagram / State Diagram**.<br>4. Identify the main states of the Women Safety Alert System.<br>5. Add the **Initial State**.<br>6. Create the **App Launched** state.<br>7. Add **Authentication** for login or registration.<br>8. Add the **Home** state after successful authentication.<br>9. Add **Location Tracking** and **Alert Triggered** states.<br>10. Add **Notification** and **Alert Logged** states.<br>11. Add **Manage Emergency Contacts** and **Logout** states.<br>12. Connect the states using arrows and label the transitions.<br>13. Add the **Final State**.<br>14. Arrange the diagram properly and save it. |
| | Code:<br>**Not Applicable.**<br>A UML Package Diagram is a graphical representation, so programming code is not required. |
| | Output:<br>  |

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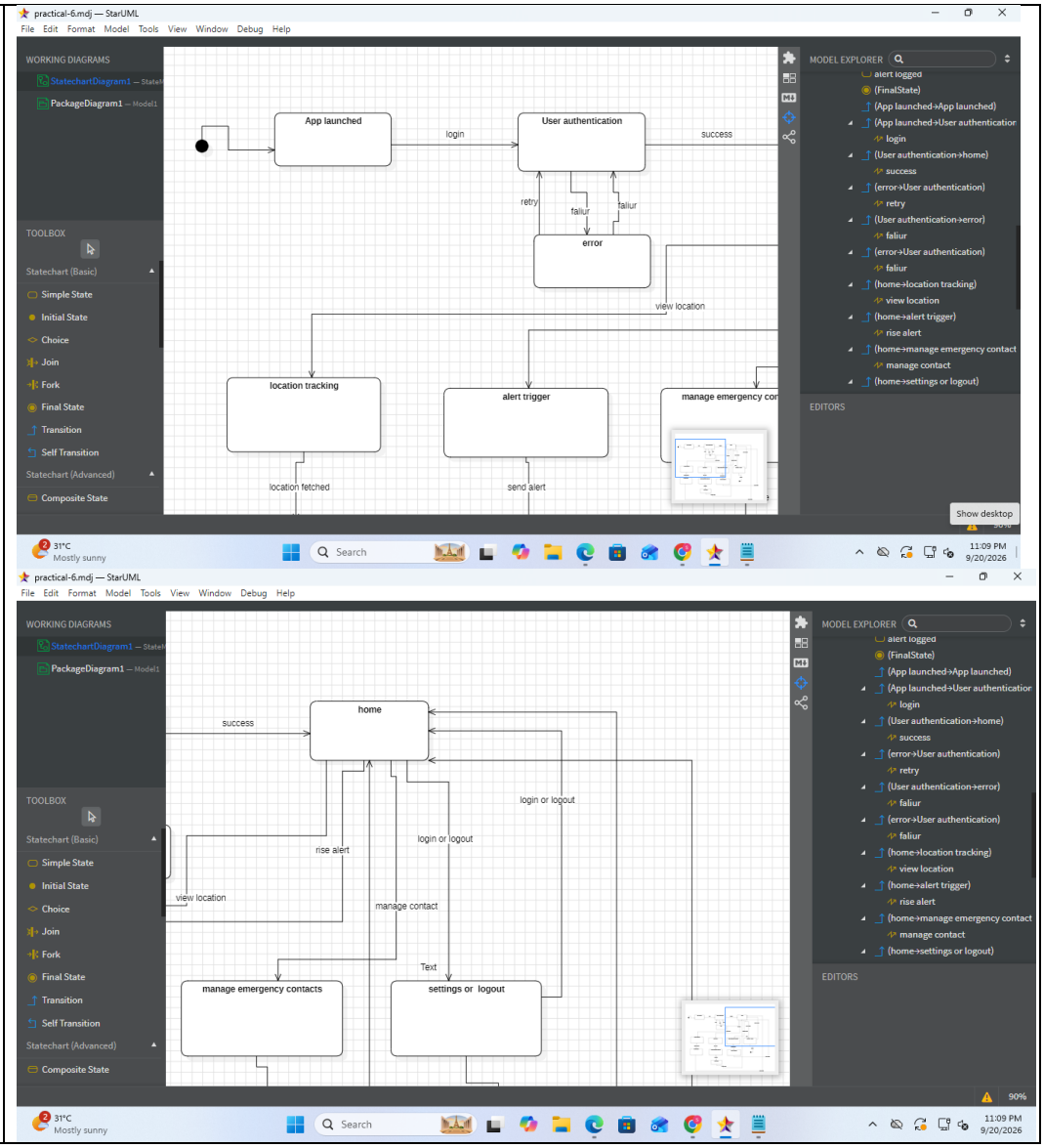
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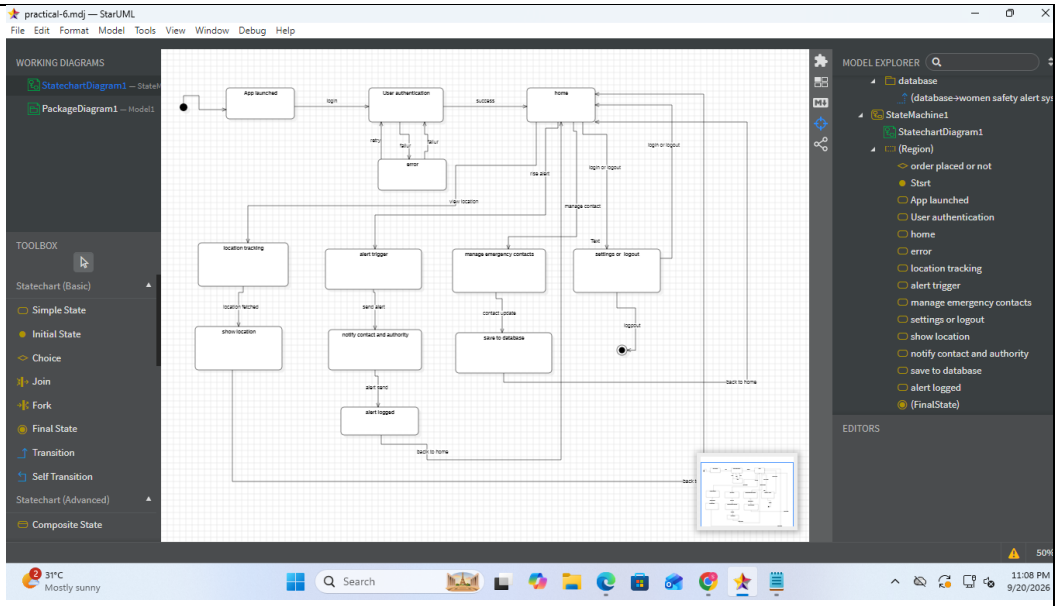


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| |   |
| Output Analysis | The State Diagram clearly shows the different states and transitions of the Women Safety Alert System. It explains how the system changes its state when the user performs actions such as login, location tracking, raising an SOS alert, managing emergency contacts, or logging out. |
| Link of student Github profile where lab assignment has been uploaded | https://github.com/SRchaudhari15/Software-Engineering-Practicals-24.git |
| Conclusion | The **State Diagram** for the Women Safety Alert System was successfully studied and drawn. It represents the different states of the system and shows how the system changes from one state to another based on user actions and events. |

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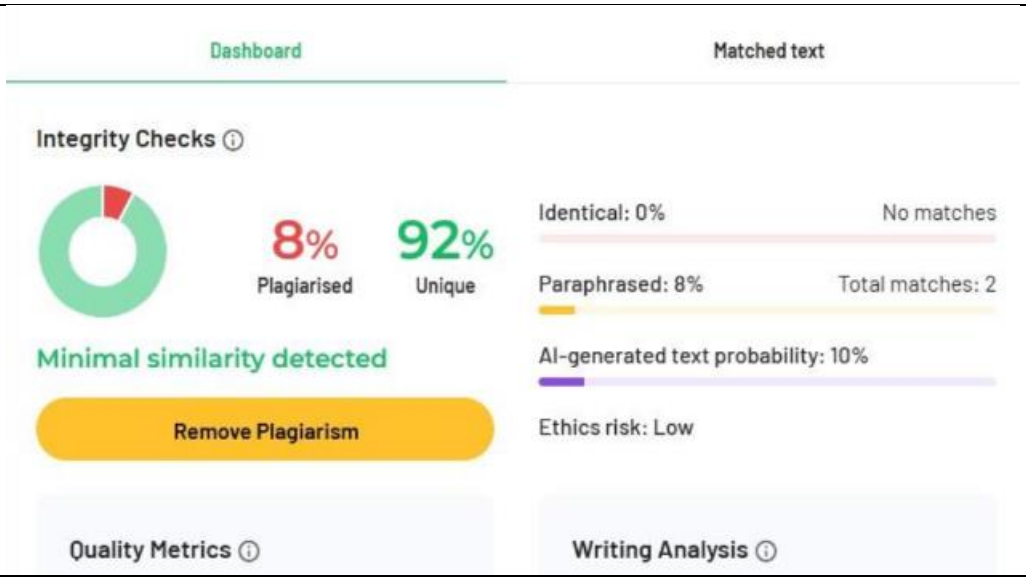
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